

$$1a) \quad |E| \Leftrightarrow \cos \ln x + \frac{\sqrt{2}}{2} \sin(\ln x^2) - \frac{1}{2} = \frac{1}{\sqrt{2}} \quad (\text{avec } m > 0)$$

$$\Leftrightarrow \cos \ln x + \frac{\sqrt{2}}{2} \sin(2 \ln x) = \frac{1+\sqrt{2}}{2}$$

$$\Leftrightarrow \cos \ln x + \sqrt{2} \sin(\ln x) (\cos \ln x) = \frac{1+\sqrt{2}}{2}$$

$$\Leftrightarrow 2 \cos \ln x \left(\frac{1}{2} \cos \ln x + \frac{\sqrt{2}}{2} \sin \ln x \right) = \frac{1+\sqrt{2}}{2}$$

$$\Leftrightarrow 2 \cos \ln x \cos(\ln x - \pi/3) = \frac{1+\sqrt{2}}{2}$$

$$\Leftrightarrow \cos(2 \ln x - \pi/3) + \cos(\pi/3) = \frac{1+\sqrt{2}}{2} \quad \text{car } 2 \cos a \cos b = \cos(a+b) + \cos(a-b)$$

$$\Leftrightarrow \cos(2 \ln x - \pi/3) = \frac{\sqrt{2}}{2} = \cos \pi/4$$

$$\Leftrightarrow \begin{cases} 2 \ln x - \pi/3 = \pi/4 + 2k\pi \\ \text{ou} \\ 2 \ln x - \pi/3 = -\pi/4 + 2k\pi \end{cases} \quad k \in \mathbb{Z}$$

$$\Leftrightarrow \begin{cases} x = e^{\frac{\pi/4 + k\pi}{2}} > 0 \quad k \in \mathbb{Z} \\ \text{ou} \\ x = e^{\frac{\pi/4 + k\pi}{2}} > 0 \end{cases} \quad \text{SIR} = \left\{ e^{\frac{\pi/4 + k\pi}{2}}, e^{\frac{\pi/4 + k\pi}{2}} \right\} \quad k \in \mathbb{Z}$$

$$b) \quad (\log_2 m)^{\log_2 m} = 1 \Leftrightarrow x^y = 1 \quad \text{ou} \quad y = \log_2 x \quad (m > 0)$$

$$\Leftrightarrow y \ln y = 0$$

$$\Leftrightarrow y = 0 \text{ ou } \ln y = 0$$

$$\Leftrightarrow \log_2 m = 0 \text{ ou } \log_2 x = 1$$

$$\text{d'où avec } y \neq 0 \quad \text{SIR} = \left\{ 2^0 = 1, 2^1 = 2 \right\}$$

$$c) \quad 16^m + 16^{1-m} = 10 \Leftrightarrow (16^m)^2 + 16 = 10 \times 16^m$$

$$\Leftrightarrow y^2 - 10y + 16 = 0 \quad \text{ou} \quad y = 16^m > 0$$

$$\Leftrightarrow y^2 - (8+2)y + 2 \times 8 = 0$$

$$\Leftrightarrow 16^m = 2 \text{ ou } 8$$

$$\text{SIR} = \left\{ \frac{\ln 2}{\ln 16} = \frac{1}{4}, \frac{\ln 8}{\ln 16} = \frac{3}{4} \right\}$$

$$3/|E| \Rightarrow \operatorname{arch}(1-x) + \operatorname{arch} x = \pi/3 \quad (x > 0)$$

$$\Rightarrow \tan(\operatorname{arch}(1-x) + \operatorname{arch} x) = \tan \pi/3 = \sqrt{3}$$

$$\Rightarrow \frac{(1-x)+x}{1-(1-x)x} = \sqrt{3} \quad \text{con } \forall a, b \in \mathbb{R}, \tan(a+b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

$$\Rightarrow x^2 - x + \frac{\sqrt{3}-1}{3} = 0$$

$$\Delta = \frac{-3+4\sqrt{3}}{3} < 0 \quad \underline{S_{\mathbb{R}^+} = \emptyset}$$

e) Interval (Polinomi 2nd degree)

$$f) \sin x + \cos x = 2 \sin x + \sin x \quad (\Rightarrow) \quad \frac{e^{ix} + e^{-ix}}{2} - 2 \left(\frac{e^{ix} - e^{-ix}}{2i} \right) = \sin x + \cos x = \sqrt{2} \sin(x - \pi/4)$$

$\alpha \in [\pi/2, \pi]$

$$\Rightarrow -e^{ix} + 3e^{-ix} = 2i$$

$$\Rightarrow (e^{ix})^2 + 2i(e^{ix}) - 3 = 0$$

$$\Delta = 4 \times 2 \sin^2(x - \pi/4) + 12 > 0$$

$$e^{ix} = \frac{-2i \sin(x - \pi/4) \pm \sqrt{3 + 2 \sin^2(x - \pi/4)}}{1}$$

$$\text{or } \forall \alpha \in \mathbb{R}, \quad 3 + 2 \sin^2(\alpha - \pi/4) > 2 \sin^2(\alpha - \pi/4) \Rightarrow \sqrt{3 + 2 \sin^2(\alpha - \pi/4)} > \sqrt{2} \sin(\alpha - \pi/4)$$

$$\Rightarrow \begin{cases} -\sqrt{2} \sin(\alpha - \pi/4) + \sqrt{3 + 2 \sin^2(\alpha - \pi/4)} > 0 \\ -\sqrt{2} \sin(\alpha - \pi/4) - \sqrt{3 + 2 \sin^2(\alpha - \pi/4)} < 0 \end{cases}$$

$$\underline{S_{\mathbb{R}} = \left\{ \ln(-\sqrt{2} \sin(\alpha - \pi/4) + \sqrt{3 + 2 \sin^2(\alpha - \pi/4)}) \right\}, \alpha \in \mathbb{R}}$$

• Si α est aussi une incognue

$$\underline{S_{\mathbb{R}^2} = \left\{ \ln(-\sqrt{2} \sin(\lambda - \pi/4) + \sqrt{3 + 2 \sin^2(\lambda - \pi/4)}) \right\}, \lambda \in \mathbb{R}}$$

$$\begin{aligned}
 g) J &= \int_0^1 x^1 \operatorname{arctanh} x \, dx \\
 &= \left[\operatorname{arctanh} x \right]_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} \, dx \\
 &= \pi/4 - \frac{1}{2} \left[\ln(1+x^2) \right]_0^1
 \end{aligned}$$

$$J = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$h) I = \int \frac{7}{x + x \ln^2 x} \, dx$$

$$= 7 \int \frac{1/x}{1 + \ln^2 x} \, dx$$

$$= 7 \operatorname{arctan} \ln x + C, \quad C \in \mathbb{R}$$